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DATA SCIENCE(24AD3107)

2026-27, ODD Semester

ABSTRACT

Sec No-08	**Team No-24**	Project Title- **Interactive EDA Dashboard** – Upload any CSV and auto-generate distribution plots, box plots, correlation heatmaps, and pairplots using Streamlit/Dash.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Name
Data Ingestion, Preprocessing & Exploratory Data Analysis Module	2420030563	B.Harneesh
Data Visualization & Interactive Streamlit Dashboard Module	2420030639	Dhanush

1. Abstract:

Exploratory Data Analysis (EDA) is one of the most important stages in the data science workflow. Before applying statistical models or machine learning algorithms, it is necessary to understand the structure, quality, distribution, and relationships within a dataset. Traditional EDA generally requires users to write multiple lines of code to inspect datasets, calculate descriptive statistics, identify missing values, detect outliers, and create different types of visualizations. This process can become repetitive and time-consuming when working with multiple datasets, particularly for beginners and users who have limited programming experience.

The Interactive EDA Dashboard is a web-based data science application developed to simplify, automate, and improve the process of exploratory data analysis. The proposed system allows users to upload a CSV file through an interactive interface and automatically performs a series of preliminary data analysis operations. Once the dataset is uploaded, the system examines its dimensions, column names, data types, missing values, duplicate records, and basic statistical characteristics. This provides users with an immediate overview of the dataset without requiring them to write separate analysis programs.

The dashboard focuses on automatically generating commonly used EDA visualizations. Distribution plots are provided to understand the distribution and spread of numerical variables, while box plots help identify the central tendency, variability, and potential outliers in the data. A correlation heatmap is generated to visualize the relationships between numerical variables and identify variables that have strong or weak linear associations. In addition, pairplots provide a comprehensive visual comparison of multiple numerical variables, helping users identify trends, clusters, relationships, and unusual observations. These visualizations allow users to explore the dataset from different perspectives and obtain meaningful insights more efficiently.

The system can be implemented using Python and commonly used data science libraries such as Pandas and NumPy for data processing and Matplotlib, Seaborn, or Plotly for data visualization. Streamlit or Dash can be used to develop the interactive web interface. The dashboard can include interactive controls that allow users to select columns, apply filters, choose visualizations, and explore different aspects of the uploaded dataset. This makes the application flexible enough to work with a variety of structured CSV datasets.

The proposed system aims to reduce repetitive coding, save analysis time, and make exploratory data analysis more accessible to students, researchers, analysts, and other users. Rather than manually creating individual plots and statistical summaries, users can obtain an initial understanding of their dataset through a single interactive platform. The dashboard can also serve as a useful preliminary analysis tool before performing advanced statistical analysis or machine learning.

Overall, the Interactive EDA Dashboard demonstrates the practical application of data science, data visualization, and web-based technologies in developing an automated analytical tool. By integrating data preprocessing, descriptive statistics, automated visualization, and user interaction into a single platform, the project provides an efficient and user-friendly approach to understanding datasets and discovering useful patterns, relationships, and anomalies within the data.

Justification:

Exploratory Data Analysis (EDA) is an essential step in the data science process because it helps users understand the structure, quality, distribution, and relationships present in a dataset before performing advanced analysis or machine learning. However, traditional EDA often requires users to repeatedly write and modify code for each dataset, making the process time-consuming and difficult for beginners.

The Interactive EDA Dashboard is proposed to address this problem by providing a single, reusable platform for performing common EDA tasks automatically. Users can upload a CSV dataset and obtain important information such as data dimensions, data types, missing values, duplicate records, and descriptive statistics without writing separate programs.

The dashboard also automatically generates important visualizations such as distribution plots, box plots, correlation heatmaps, and pairplots. These visualizations help users understand data distributions, identify potential outliers, examine relationships between variables, and discover patterns within the dataset. Providing these features in one platform reduces repetitive coding and makes the analysis process faster and more convenient.

The project is also useful for students, researchers, and beginner data analysts because it provides an interactive and visually understandable approach to EDA. Instead of focusing on writing code for every analysis task, users can focus on interpreting the results and gaining meaningful insights from the data.
From a technical perspective, the project provides practical implementation of important data science concepts such as data ingestion, preprocessing, descriptive statistics, data visualization, and interactive dashboard development. Technologies such as Python, Pandas, NumPy, Seaborn, Matplotlib, and Streamlit can be integrated to create an efficient and user-friendly system.

Therefore, the project is justified as a practical solution for reducing manual EDA effort, saving analysis time, improving data understanding, and providing a reusable interactive platform for preliminary exploration of structured datasets.

Guide Name and Signature (remarks if any) : Dr. N. Sunanda